

Introduction to Bioinformatics

Microarray, NGS

Ali Sharifi-Zarchi Somayyeh Koohi

CE Department
Sharif University of Technology

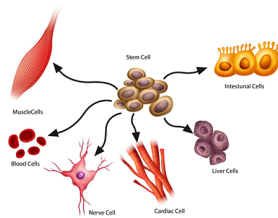
November 15, 2025

Biological Diversity, Differentiation, and Disease



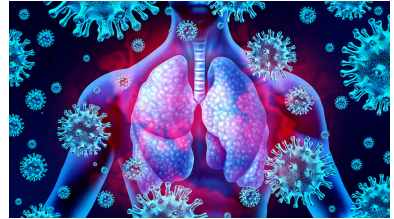
Biodiversity

The vast variety of life forms — from microbes to mammals — shaped by evolution and adaptation to different environments.



Differentiation

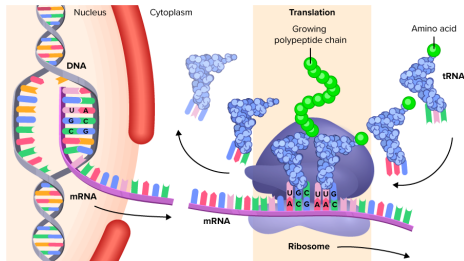
The process by which cells with the same genome acquire distinct structures and functions through regulated gene expression.



Disease

Alterations in normal biological processes that disrupt gene regulation and cellular balance, often leading to pathology.

What to Read & How to Read



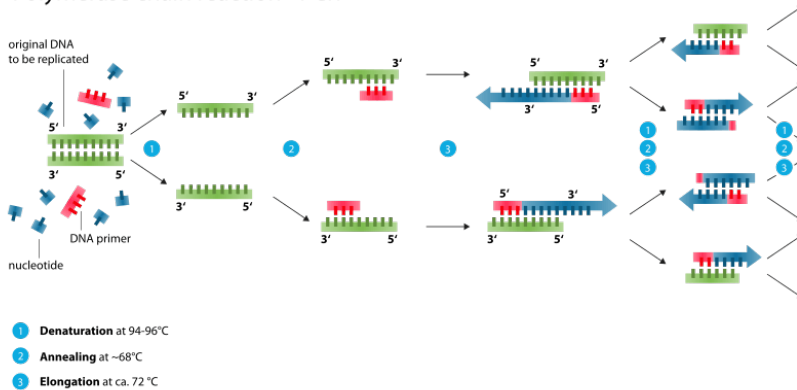
What to Read & How to Read

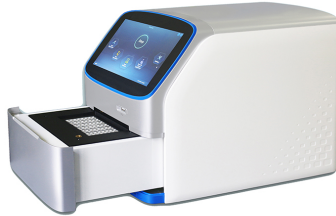
- The genetic information stored in DNA is transcribed into different types of RNA — mRNA, tRNA, rRNA, and non-coding RNAs — each carrying specific biological roles.
- Reading involves decoding the RNA or DNA sequences to interpret which genes are active and how they contribute to cellular processes.



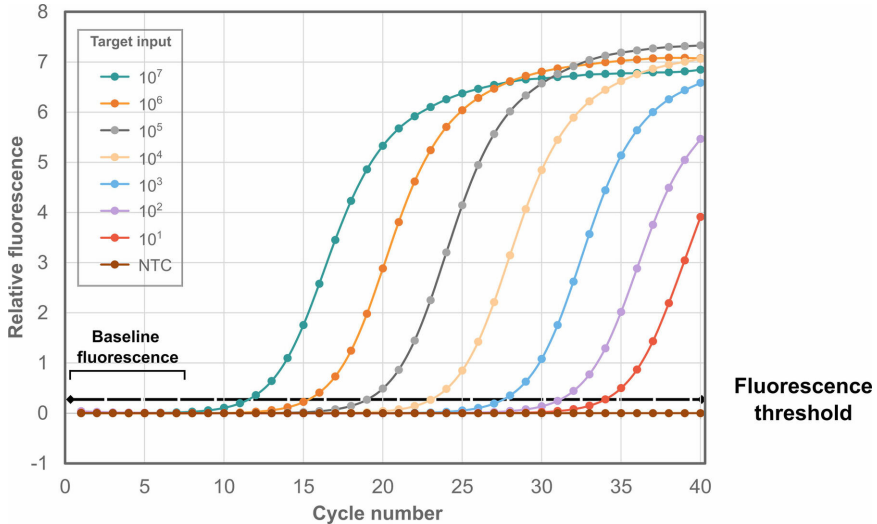
- **Purpose:** Amplifies specific DNA sequences exponentially from small starting amounts.
- **Principle:** Repeated cycles of denaturation, primer annealing, and DNA synthesis by a thermostable DNA polymerase.
- **Output:** Produces millions of copies of target DNA, but only detectable at the end (endpoint analysis).
- **Applications:** Used for cloning, genotyping, and detecting presence or absence of DNA sequences.

Polymerase chain reaction - PCR

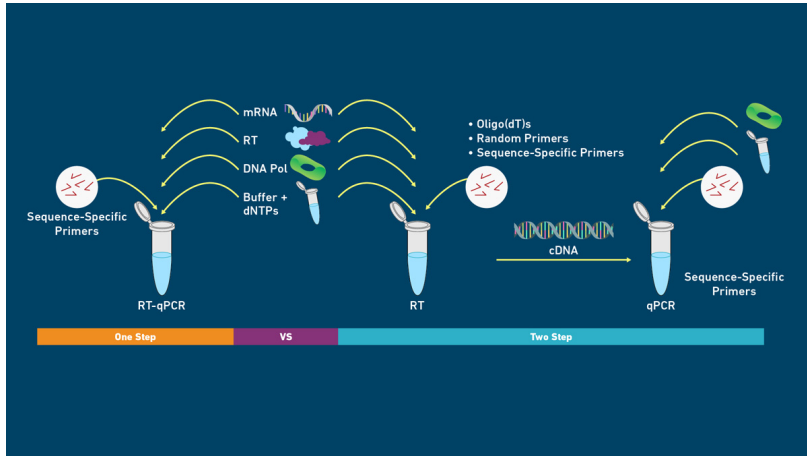




- **Purpose:** Measures DNA amplification in real time during the PCR reaction.
- **Principle:** Uses fluorescent dyes (e.g., SYBR Green) or probes (e.g., TaqMan) to monitor DNA synthesis each cycle.
- **Output:** Generates amplification curves and Ct (threshold cycle) values for precise DNA quantification.
- **Applications:** Quantifying DNA copy number, pathogen detection, and mutation analysis.

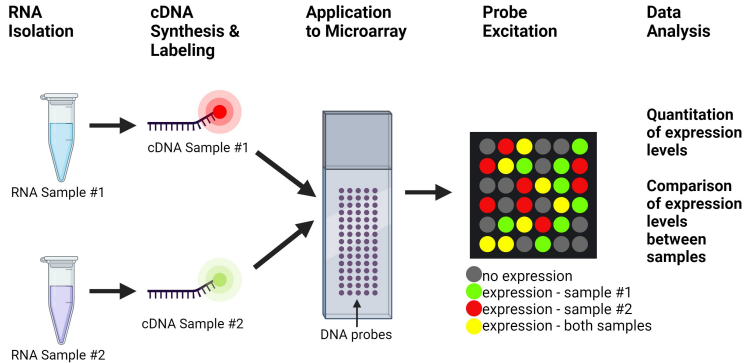


qRT-PCR

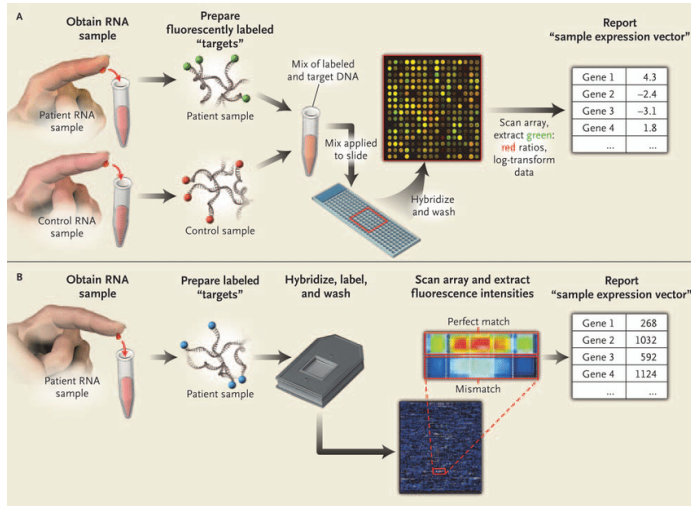


- **Purpose:** Quantifies RNA expression by converting RNA into complementary DNA (cDNA) before qPCR.
- **Principle:** Combines reverse transcription (RT) and real-time qPCR in one or two steps.
- **Output:** Measures relative or absolute gene expression levels through mRNA quantitation.
- **Applications:** Studying gene expression, validating RNA-Seq data, and detecting RNA viruses (e.g., SARS-CoV-2).

Microarray



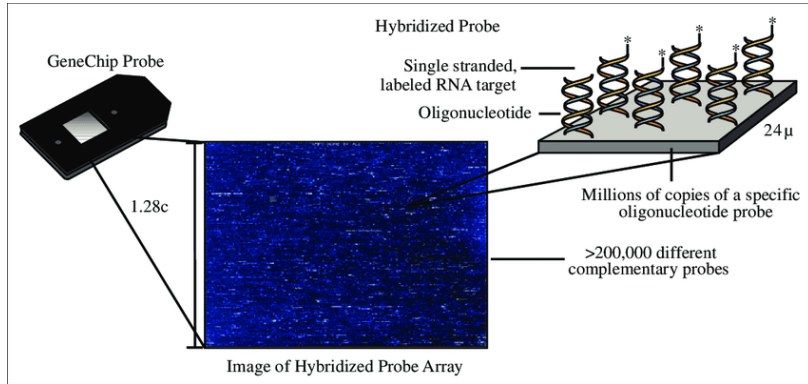
Microarray



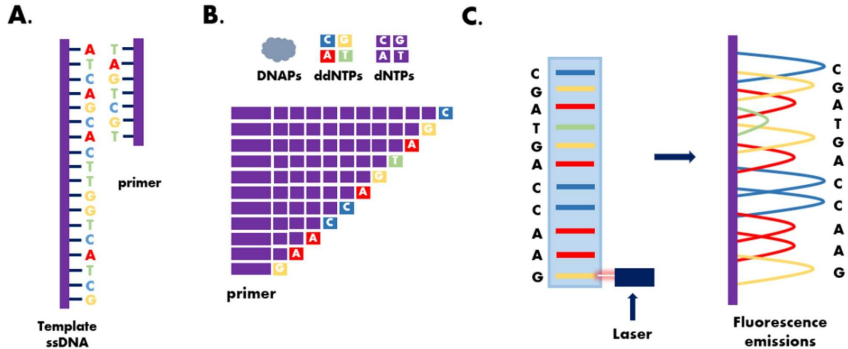
Microarray

- Microarrays measure expression levels of thousands of genes simultaneously.
- DNA probes fixed on a solid surface hybridize with fluorescently labeled cDNA.
- Used in genomics for profiling gene activity, mutations, or polymorphisms.

Microarray



Sequencing – Sanger



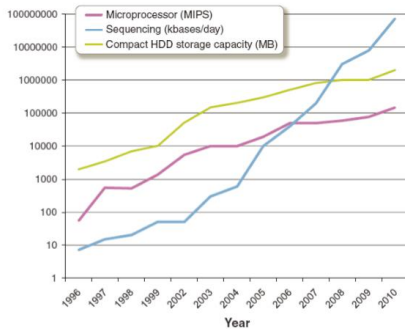
Sequencing Method: Sanger

- **Principle:** Based on selective incorporation of chain-terminating dideoxynucleotides (ddNTPs) during DNA replication.
- **Process:**
 - DNA polymerase synthesizes a complementary strand from a single-stranded DNA template.
 - Incorporation of a ddNTP stops elongation at specific bases.
 - The resulting fragments of varying lengths are separated by electrophoresis.
- **Detection:** Fluorescent labels on ddNTPs reveal the terminal base at each position, allowing the sequence to be read.
- **Features:** High accuracy but low throughput — suitable for small-scale sequencing projects and validation studies.

Sequencing



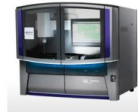
Next Generation Sequencing



Roche/454



Illumina HiSeq



SOLiD 5500



Ion Proton

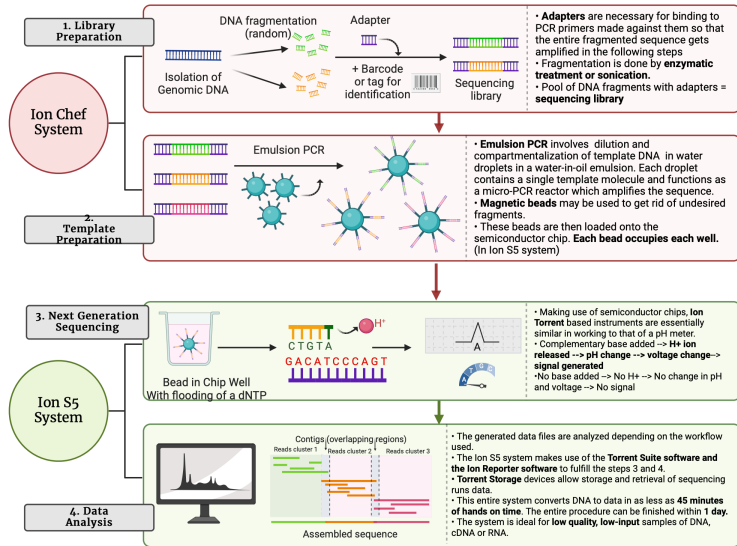


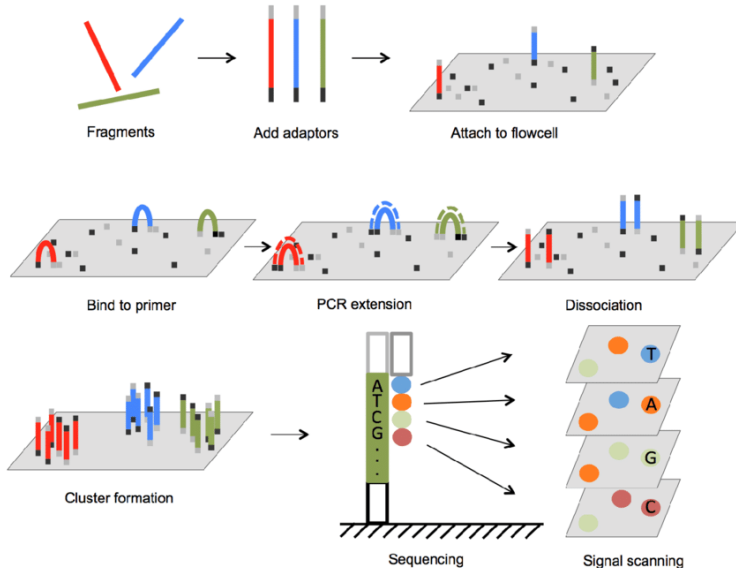
PacBio RS



Oxford Nanopore

Ion Torrent





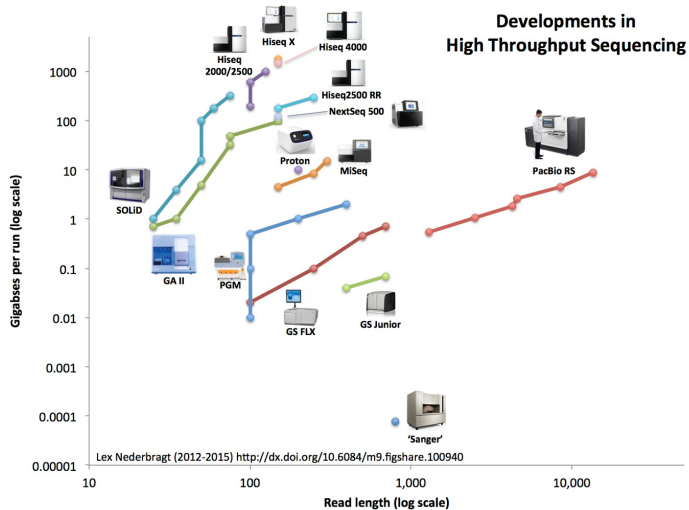
Sequencing Method: Illumina

- **Principle:** Sequencing by synthesis (SBS) — detects fluorescently labeled nucleotides as they are incorporated into growing DNA strands.
- **Process:**
 - DNA fragments are attached to a flow cell and amplified to form clusters.
 - During each cycle, a single fluorescently labeled base is added, imaged, and then chemically cleaved to allow the next base incorporation.
- **Detection:** A high-resolution camera captures color signals to determine the base added at each cluster position.
- **Features:** Extremely high throughput, short read lengths, and low per-base cost — ideal for whole-genome, transcriptome, and targeted sequencing.

Illumina Sequencing — Step-by-step

- **Extraction** — isolate DNA/RNA from the sample.
- **Library prep** — fragment nucleic acids and add adapters.
- **Amplification/cleanup** — enrich and purify the library.
- **Cluster generation** — attach library to flow cell and amplify.
- **Sequencing** — read bases through sequencing-by-synthesis.
- **Base calling & demultiplexing** — convert signals to reads and separate samples.
- **Alignment & analysis** — map reads and perform downstream QC/interpretation.
- **Video explanation Link**

NGS Throughput



NGS Throughput

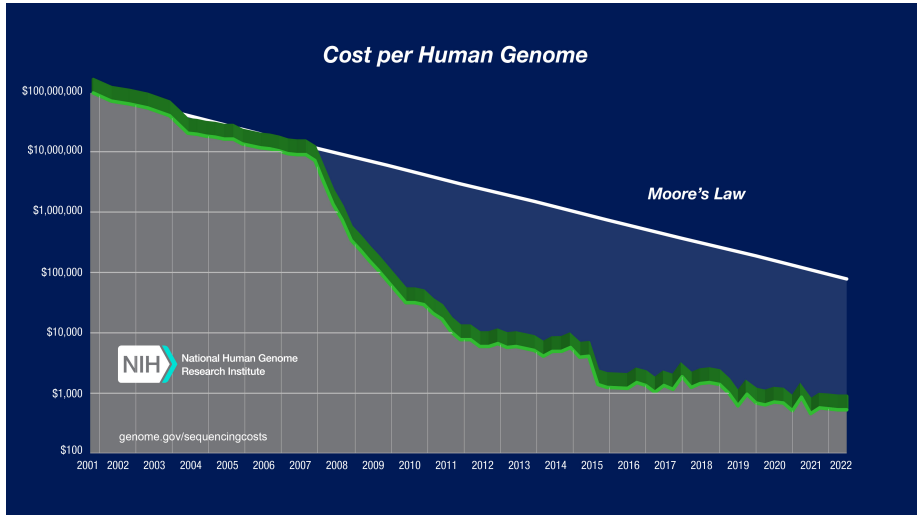
24 Hours



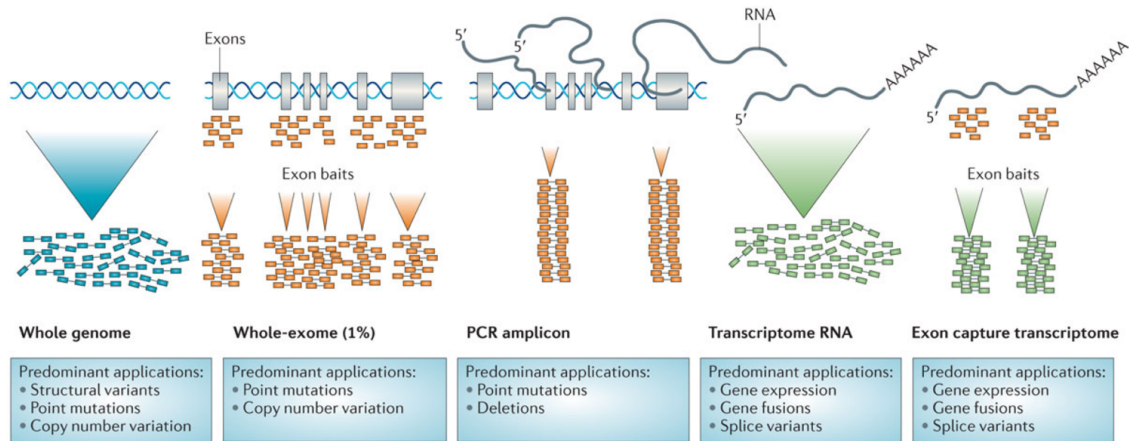
709 Years



NGS Cost



Applications



Core Sequencing Methodologies

- **Single-End**

- Reads fragments from **one side only**.
- **Use:** Fast and cost-effective for applications like gene expression counting.

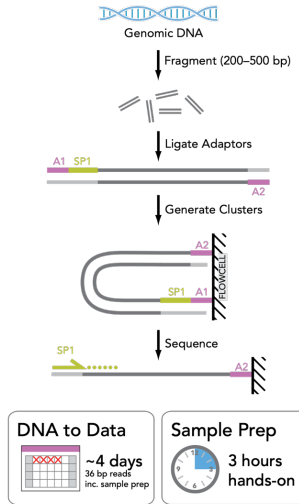
- **Paired-End**

- Reads the **same fragment** from **both sides**.
- **Use:** Massively improves alignment accuracy and helps find small structural variants (indels).

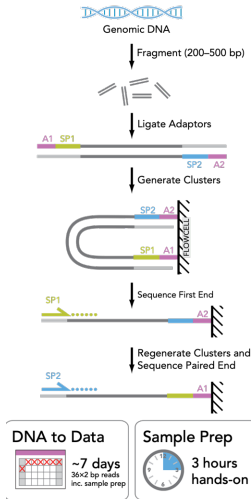
- **Mate-Pair**

- Reads two ends of a **very long fragment** (kilobases apart).
- **Use:** Essential for *de novo* genome assembly and detecting large structural rearrangements.

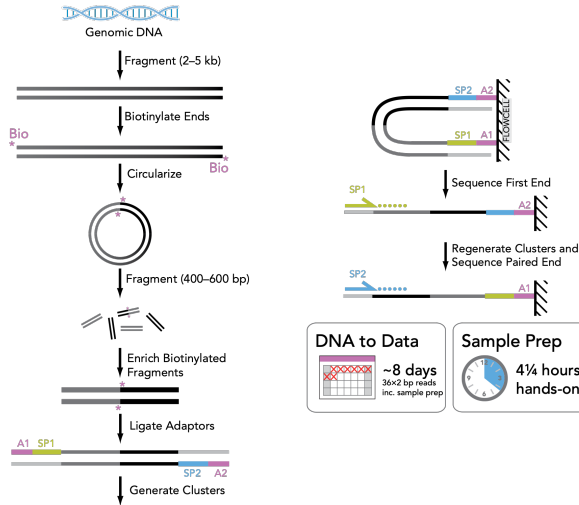
Single-end



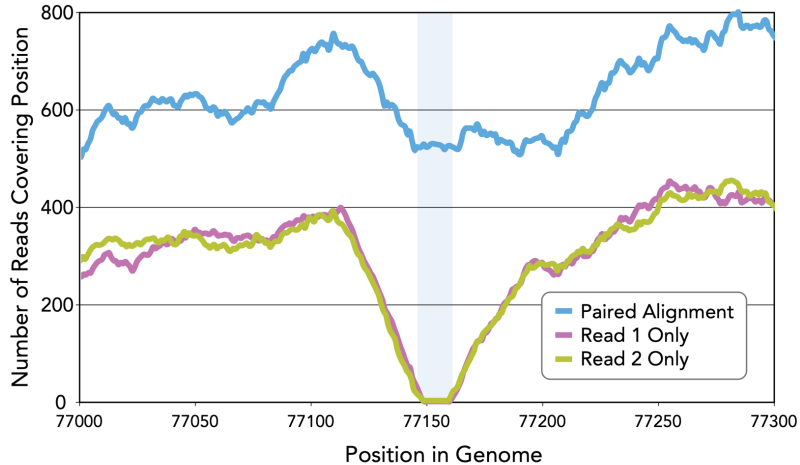
Paired-end



Mate-pairs



Paired-end vs. Single-end



Yasamin Tafakor
Raoufe Rezai

References

- Curry, E. *Introduction to Bioinformatics with R*.
- Istrail, S., & Pevzner, P. *An Introduction to Bioinformatics Algorithms*.